

GPU Computing

Exercise 2

Luis Wenzel
Miro Joensuu
Daniel Bayer
Noah Schneider

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1 Presenting

Willing to present:

6.1.1 Yes

6.1.2 Yes

2 Reading

2.1 Debunking the 100X GPU vs. CPU Myth: An Evaluation of Throughput Computing on CPU and GPU

This article successfully shows that GPUs are not unlike advertised 100-1000x faster than CPUs but more in the ballpark of 2.5% faster. This was done by systematically comparing different throughput-kernels and optimizing them for GPU as well as CPU. The paper therefore showed how important fair treatment optimizations is to get scientific results.

The misconception around the alleged advantage of GPU-performance originated in the unfair comparison of underpowered and not optimized CPU code with highly GPU-programs. But actually both CPU and GPU can profit from platform specific optimizations, which closes the gap between their performance. Also the biggest performance difference does not originate on the raw compute power of these devices but more storage-speed, cache-size, data access patterns as well as the cost of synchronization.

In our opinion this article is very important because it aligns the expectations of GPU-computing more with the reality of the actual expected performance. The methodology seems fair, even though the researchers are from Intel they did optimizations of the GPU as well. Altogether this article shows nicely that it is more important to optimize for the architecture you work with than the architecture itself.

2.2 NVIDIA Tesla: A unified graphics and computing architecture

This article explains how GPU-performance took a big leap with NVIDIA's Tesla-architecture by joining the processors of vertex- and pixel-processing to a single unified processor architecture design. By combining these programs to the same core, it enabled dynamic load management and therefore increased processor utilization compared to the fixed ratios. Also Tesla enabled general-purpose-computing with a GPU (GPGPU) by creating the powerful parallel-computing-platform called Compute Unified Device Architecture (CUDA) which is programmable in C.

The traditional pipeline with fixed ratios of pixel- and vertex processors is rarely balanced because the size of triangles are largely different and therefore some core sit still waiting for new tasks while the others are over-utilized. By combining these processes to a unified processor design (Streaming Multiprocessor) it not only allows for better load balancing but also lowers cost and the complexity of the chip-design by having less specialized hardware. Tesla also implemented the Single-Instruction-Multiple-Thread (SIMT)-model the first time. This clusters independent threads into warps which allows a more efficient execution of branched code.

In our opinion the Tesla architecture is a historic milestone of universal usage of GPUs for high-performance computing by providing more generalized core with the necessary interface (CUDA) to program them efficiently. These concepts are still in use in every modern NVIDIA-GPU and consolidates their marked dominance in AI and HPC.